

RSA Algorithm.

what are ^{the} prime factors

For 35

$$7 \times 5 = 35$$

7 and 5

what are the prime factors
of 250 digit decimal number

Steps:-

1) How to Generate public
key and private key

2) How to encrypt using
Public Key

3) How to decrypt using
private key.

Rules for generating a public
key

→ Select two prime numbers

p and q

$$\text{Let } n = p \times q$$

→ Select a number e such that

$$e \text{ is } 1 < e < \phi(p) \phi(q) \text{ and}$$

$$\text{gcd}(e, \phi(p) \phi(q)) = 1$$

The public key is $\{e, n\}$

Public key example

$$p = 3 \text{ and } q = 11$$

$$n = p \times q \text{ i.e. } n = 3 \times 11 = 33$$

Select a number e such that

$$e \text{ is } 1 < e < \phi(p) \phi(q) \text{ and}$$

$$\text{gcd}(e, \phi(p) \phi(q)) = 1$$

$$\phi(p) \phi(q) = \phi(3) \phi(11) = 2 \times 10 = 20$$

$$\text{Let } e = 13 \text{ such that } \text{gcd}(13, 20) = 1$$

The public key is $\{e, n\}$ i.e.
 $\{13, 33\}$

Private Key generation

find a number d such that

$$e * d \bmod \phi(p) \phi(q) = 1$$

d is the modular multiplicative

inverse here

when you will have

multiplicative inverse for

a number in modular

arithmetic

$$e * d \bmod n = 1$$

only when e and n are

co-prime.

The private key (d, n)

find d .

$$e * d \bmod \phi(p) \phi(q) = 1$$

$$13 * 17 \bmod 20 = 1$$

$$d = 17$$

The private key is $(17, 33)$

Encryption.

User B want to send a message to user A

User B knows the public key of A $\{e, n\}$ i.e. $\{13, 33\}$.

User B send 4 to user A

$p = 4$ { important $p < n$ }

Encryption

$$\text{cipher} = p^e \bmod n$$

$$= 4^{13} \bmod 33 = 31$$

Decryption

Private key of user A $\{d, n\}$
that is $\{17, 33\}$

plaintext: $B^d \bmod n$

$$= 31^{17} \bmod 33$$

$$= \underline{4}$$

As a challenge,

base logarithms.

$$1) 2^5 = 32.$$

$$2) 3^? = 129140163$$

Answer - 17

We can also write.

$$5 = \log_2(32)$$

$$17 = \log_3(129140163)$$

In general:

$$p^i = Q$$

$$i = \log_p(Q) \quad \text{log of } Q \text{ base } p$$

P

Modular Arithmetic

$$9 = 3^4 \pmod{12}$$

$$26067 = 14^4 \pmod{29867}$$

$$26067 = 14^{12} \pmod{29867}$$

$$4 = d \log_{3,12} (a)$$

$$12 = d \log_{14,29867} (26067)$$

In general

$$a = b^i \pmod{m}$$

$$i = d \log_{b,m} (a)$$